Astronomy & Astrophysics manuscript no. ms
February 8, 2023
©ESO 2023

Upper Limits on CN from Falling Evaporating Bodies seen towards Beta Pictoris

M.A. Kenworthy[1], E. de Mooij[2], C. Opitom[3], A. Brandeker[4], F. Kiefer[5], and A. Fitzsimmons[1]

[1] Leiden Observatory, Leiden University, P.O. Box 9513, 2300 RA Leiden, The Netherlands
e-mail: kenworthy@strw.leidenuniv.nl
[2] Astrophysics Research Centre, School of Mathematics and Physics, Queens University Belfast, BT7 1NN Belfast, UK
[3] Institute for Astronomy, University of Edinburgh, Royal Observatory, Edinburgh EH9 3HJ, UK
[4] Institutionen för astronomi, Stockholms universitet, AlbaNova universitetscentrum, 106 91, Stockholm, Sweden
[5] LESIA, Observatoire de Paris, Université PSL, CNRS, 5 Place Jules Janssen, 92190 Meudon, France

Received XXXX; Accepted XXXX

ABSTRACT

Context. Beta Pic has Falling Evaporating Bodies (FEBs) which are seen as variable transient absorption features towards Beta Pictoris.
Aims. To detect cyanogen (CN) in the FEBs spectrum, confirming that they are similar to comets in the Solar System.
Methods. We combine the High Accuracy Radial Velocity Planet Searcher (HARPS) spectra from the past 20 years to make a high signal to noise ratio spectrum and look for the CN band head. We also combine Ultra-violet Visible Echelle Spectrograph (UVES) data to confirm our non-detections/detections.
Results. No CN is seen to a level of XXX ppt, indicating that there is less CN in these FEBs than Solar system comets.
Conclusions. FEBs are not Solar System comets in composition.

Key words. comets — transits, spectroscopy

1. Introduction

The formation and evolution of planets and their attendant moons is thought to occur in the first few ten million years of the stellar system REF. Comets deliver volatiles across menay different distances within the Solar system, from the outermost reaches of the Oort cloud in to the terrestrial planets. It is therefore important to understand the distribution, nature and chemical composition of exocomets in other planet forming systems.

One of the closest young systems is Beta Pictoris, which has been studied intensively both with direct imaging and with spectroscopy.

Beta Pictoris is a young, early type star whose stellar chromosphreic lines show rotational broadening of 200 km/s REF. Additional, much narrower absorption features are seen in the Calcium H and K lines. These features appear and disappear on several hour long timescales, and show a different range of morphologies [REF]. Intensive spectroscopic monitoring over several hours show changes of radial velocity of these components consistent with elliptical orbits. The absorption is hypothesised to be due to the evaporation of infalling bodies that pass within a few stellar radii of the star, resulting in sublimation of metals that produce a large coma containing the Calcium atoms which are then seen as an additional absorption component [REF REF].

From the multiple detections of exocomets in spectroscopy towards Beta Pictoris, the broadband detection of exocomet tails transiting a star were seen initially detected towards the star KIC 3542116 (Rappaport et al. 2018) with the Kepler satellite [REF] and subsequently detected in TESS data towards Beta Pictoris (Zieba et al. 2019). Observations of Beta Pictoris in subsequent TESS sector revealed over two dozen transits whost depth follows a power law distribution similar to those of the Solar system (Lecavelier des Etangs et al. 2022; Pavlenko et al. 2022).

To the contrary of what is observed in the case of FEBs, most molecular and atomic signatures detected in the coma of Solar system comets are emission lines/bands instead of absorptions.

Spectroscopic observations of Solar system comets when they are very close to the Sun are rare due to observational difficulties. For this comet, it was noticed that the main emission bands/lines detected in its optical spectrum dramatically changed while the comets approached the Sun. At distances greater than 0.4 au, CH, CN, C_2, and C_3 emission bands dominated the spectrum. However, at 0.15 au from the Sun, the only molecular emission detected was CN, while numerous atomic lines were visible ([O] I, Na I, K I, Ca II, Cr I, Mn I, Fe I, Ni I, Cu I, V I; Dufay et al. 1965; Thackeray et al. 1966; Preston 1967; Slaughter 1969). Additionally, a tail of Fe I atoms was proposed to explain a linear feature imaged in the tail of sungrazing comet C/2016 P1 McNaught 2006P1 (Fulle et al. 2007).

On the other hand, Na I D-line emission has been observed in many comets within $r \sim 1$ au of the Sun, where either the emission was much brighter than the foreground sky emission, or the comet was observed at high spectral resolution enabling separation from the telluric lines REFs. The high efficiency of the Na I D transition makes it easily detectable, even for relatively low column density. A notable discovery was a spectacular tail of Na I associated with comet C/1995 O1 Hale-Bopp at $r \sim 1$ au, formed by radiation pressure on the neutral atoms before ionisation (Cremonese et al. 1997).

These characteristics imply that Ca II and Fe I are predominantly locked up within cometary dust gains, and only observed when a comet is close enough to the Sun to allow their sublima-

tion. This occurs at $r \leq 0.03 - 0.06$ au for a dust grain with Bond albedo ~ 0.01 and a sublimation temperature of $T \simeq 1600$K, depending on the thermal model assumed. On the other hand, the origin of the Na I tail observed in numerous comets at larger distances from the Sun is still not very well understood (Cremonese et al. 2002). Release from dust grains as well as molecular processes have been suggested as possible sources of Na I in the coma and tail of comets annd explain its presence at large solar distances (Cremonese et al. 1997).

For the majority of Solar system comets, their optical spectra are dominated by the reflected sunlight from dust grains in their comae, plus molecular and atomic emission lines (e.g. Hyland et al. 2019). Most of the neutral and ionic molecular emission results from resonance fluorescence with solar photons. Some species, such as H_2O^+ and CO^+ result from ionization of parent molecules (i.e. molecules released directly by the sublimation of nuclear ices), all other molecules observed are second generation species, often created via photodissociation of parent molecules or released by organic-rich grains in the coma. Another process also observed in optical spectrum of comets is prompt atomic emission ([O I], [C I], and [N I]). Those forbidden transitions are the result of atoms produced in an excited state by the photodissociation of parent molecules in the coma.

Of all cometary gas emission, the highest signal to noise normally occurs with the CN (0-0) band at $\lambda \simeq 3880$Å. It is usually the first emission detected in the optical spectrum of a comet as it approaches the Sun, and as been detected in the coma of comets observed at relatively large distances from the Sun (e.g. Fitzsimmons & Cartwright 1996). The relative abundance in the coma of most comets of CN/OH$\simeq$ 500 (A'Hearn et al. 1995), together with its high fluorescence scattering efficiency factor of ~ 2.6 photons/sec/molecule at 1 au from the Sun and $\dot{r} \simeq 0$ (Schleicher 2010), gives rise to a clear signature of a cometary gas comae. Historically, the CN was believed to be created from photodissociation of HCN, which is present in abundances of the order of 1% relative to water in cometary ices. High-spatial resolution sub-mm mapping of HCN distributions supports a nuclear source in some comets (Cordiner et al. 2014). However, it is now known that in many comets a substantial fraction of CN is released from sub-micron dust grains in the inner coma of comets (e.g. Fray et al. 2005).

The C_2 (0-0) band at $\lambda \simeq 5160$Å may be of comparable flux, but has a lower contrast with the underlying dust grain continuum due to its wider bandwidth. The OH (0-0) band at $\lambda \simeq 3080$ Å generally has higher flux, but unless observed from space the high terrestrial atmospheric absorption severely diminishes the observed flux. Strong CO^+ emission bands have also been observed in a number of comets (the strongest ones are the (3-0) and (2-0) $A^2\Pi - X^2\Sigma$) REFS). However this is normally significantly weaker than the observed CN emission, with only number of comets comets.

Given the ubiquitous signature of CN emission in Solar system comets, we have embarked on an archival search for this species in the FEBs around Beta Pic. Detection of this species would provide a solid link between cometary bodies in our Solar system and the Beta Pictoris system.

2. Data description and analysis

For the study in this paper, we use all the publicly available data from the HARPS and the ESO 3.6m telescope in Chile. In addition, we complement this with observations with the **UVES!** (**UVES!**) at the Very Large Telescope in Chile.

2.1. HARPS observations

We collated all publicly available HARPS observations of Beta Pictoris obtained between October 2003 and March 2020. Although these observations cover a large baseline, the cadence between visits varies between 1 day and ~2.2 years. Similarly, the number of observations per epoch spans a wide range, between 1 and 356 exposures, while the exposure times vary between 12 and 900 seconds. This results in a SNR range per exposure between NNNN and MMMM, and a nightly SNR between NNNN and MMMM. To avoid potential edge effects from order merging, we used the E2DS files downloaded directly from the ESO archive for our analysis.

Q: Do we want to put an observing log in the appendix?

2.2. UVES observations

To supplement the HARPS observations, we use 160 epochs of UVES observations obtained between 01 April, 2017 and April 18, 2018 under programme 599.C-0751. The details for these observations are given in van Sluijs et al. (2019), and summarised here. These observations were all obtained using the same instrument setup. We used the standard 437+760 mode of UVES, which uses the #2 dichroic to allow simultaneous observations in both the red and blue arm. In this mode, the CD#2 and CD#4 cross-dispersers are used in the blue and red-arm, respectively, providing a simultaneous wavelength coverage of ~3730 to ~5000 in the blue arm, and ~5650 to ~9560 in the red arm. For the remainder of the paper, we will only focus on the data from the blue arm of UVES, as this covers the strong CN (0-0) band-head seen in Solar system comets. For the blue arm, a 0.3"-wide slit was used to obtain the highest possible resolving power (~90,000 before an intervention on the instrument in October 2017, and ~100,000 after the intervention).

The data were reduced using the ESO **UVES!** pipeline version 6.1.6 through ESO Reflex tool in order to provide as homogeneous a dataset as possible. The flat-field was applied on a pixel-by-pixel basis.The pipeline was set to output the intermediate products and the blaze-corrected spectra before merging were used for our analysis, as the order-merging could introduce systematics in the overlap region. Optimal extraction was performed, using an empirical profile

2.3. Initial data processing

Starting from the 2d spectra, the data from both instruments was processed in similar ways. Each order was treated separately, with the main focus on orders NN, MM and OO for HARPS and NN, MM, OO for UVES as those orders contain the main CN bandhead as well as the Ca II H & K lines, which are used to identify the exocomets. All the spectra were transformed into the stellar rest-frame and onto a common wavelength grid using linear interpolation.

The number of observations varies from night to night, from a single spectrum to over three hundred spectra. An average of all the spectra in one night was made for each of the nights of observations, after correcting for variations in the effective blaze profile (e.g., due to instrumental changes, differential slit losses, atmospheric absorption etc.), as outlined below. Note that after this correction, the HARPS data still have the overall instrumental blaze-profile, while the 2d-spectra from UVES were already corrected for this.

2.3.1. Effective Blaze function correction

One of the nights with a high signal to noise was chosen to act as an initial estimate for the blaze function (the night of DATE for HARPS and DATE for UVES). The individual spectra in this night were averaged together to further boost the signal-to-noise ratio. Subsequently, each individual spectrum was divided by this reference spectrum to highlight the changes in the effective profile. This ratio spectrum was subsequently binned using a bin-width of 101 pixels using a robust algorithm to remove outliers within each bin. This binned ratio was than fit using a second order polynomial. The original spectrum was finally divided by this polynomial to correct for the changes.

2.3.2. Combining the spectra

After the blaze-variation corrections, all the spectra within a night are combined in order to further boost the signal-to-noise and improve the detection limits. This could potentially re For all the night spectra:

We median combine all the nights together.

Bin the spectrum and clip it, interpolate using a cubic spline.

Divide this interpolated cubic spline into each of the nightly spectra.

THis removes the blaze function, but there remain low spatial frequency variations between different nights, due to the changing slit illumination as the star moves across the spit aperture.

We then fit a second order polynomial to each individual night and divide this out, resulting in a flattened spectrum for each night.

Subsequently, we applied a correction for variations in the instrumental blaze by selecting a reference spectrum and dividing all the spectra by this reference. The ratio was then binned, removing outliers, and a polynomial was fit to the data. For UVES we used a Nth degree polynomial, while for HARPS we used a 2nd degree polynomial. This polynomial was then divided from the spectrum under consideration. We note that the blaze correction will also remove non-instrumental effects, including variations in atmospheric transmission and seeing.

The blaze-corrected spectra taken within a night were averaged together [Should we use a weighted average?]. We note that for HARPS, the 2d-unmerged spectra still contain the overal instrumental blaze-profile, which is (partially) corrected for by using a dedicated blaze-frame obtained from the archive, which removes most, but not all, of this profile.

3. Searches for cometary species

After the initial data analysis, we proceed to with our search for lines from CN. Although we are interested in lines from the exocomets, we first coadd the data from all the individual epochs, in the stellar rest-frame, to obtain a deep search for circumstellar CN absorption. These coadded spectra for HARPS and UVES are shown in Figure. No obvious feature jumps out, and we can rule out circumstellar CN down to NNNN.

Since the exocomets have a clear radial velocity with respect to Beta Pictoris, and since each exocomet will have its own distinct velocity, we selected the NN epochs with the strongest exocometary features, a seen in the Ca II H & K lines, as well as the MM epochs with the weakest cometary features. The selection was done by

The spectra with only weak exocometary features were co-averaged in the stellar restframe in order to create a stellar master spectrum. This master-spectrum was divided out of all the remaining spectra in order to create relative spectra. The relative velocities of the exocomets were determined by ....., and the spectra were shifted to the rest-frames of these comments, before being combined. The combined exocomet aligned spectra for both the CA H&K lines and CN band are shown in Fig. **??**. As can be seen no signal is visibly present. [Do we need to add a correction to account for any radiation pressure, if so, is there a way to estimate that?]

[Do we want to do a simple CCF to see if we can combine the individual lines?]

We have theoretical specrtra of CN shown in Figure 1.

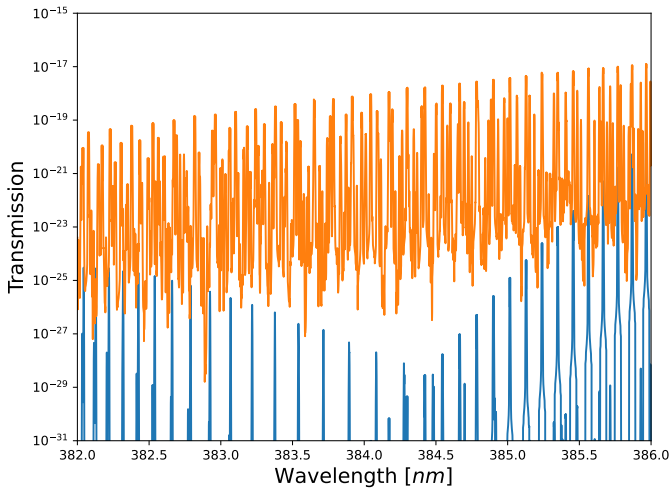

Fig. 1. Theoretical cross section spectra of CN for 100K and 1000K.

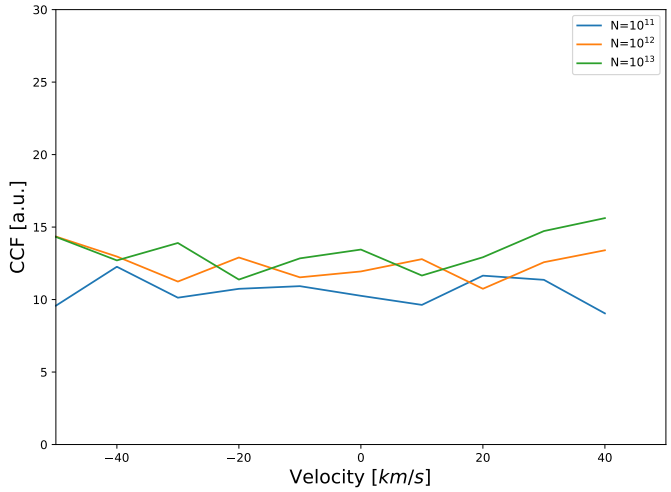

Fig. 2. Cross correlation for different number densities of CN.

3.1. CN

3.2. Other Species? CO^+, C_2?

4. Discussion

5. Conclusions

Acknowledgements. The authors thank the Lorentz Centre at Leiden University for organising the workshop "Exocomets: Understanding the Composition of Planetary Building Blocks" which led to the study in this paper. Part of this work was supported by the German *Deutsche Forschungsgemeinschaft, DFG* project number Ts 17/2–1. We made use of the `Python` programming language

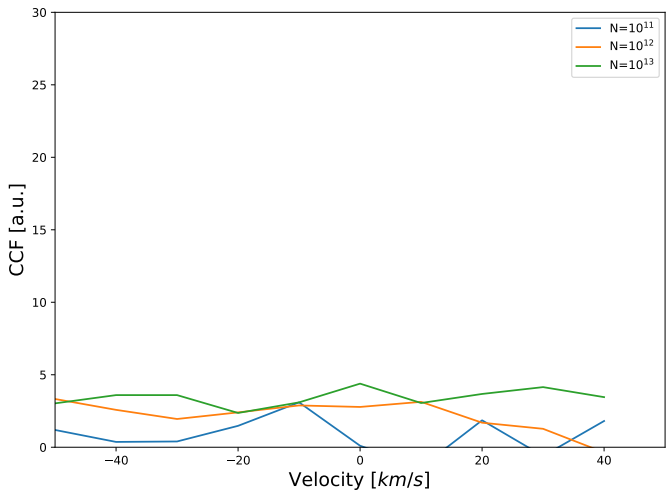

Fig. 3. Cross correlation for different number densities of CN averaged over many spectra.

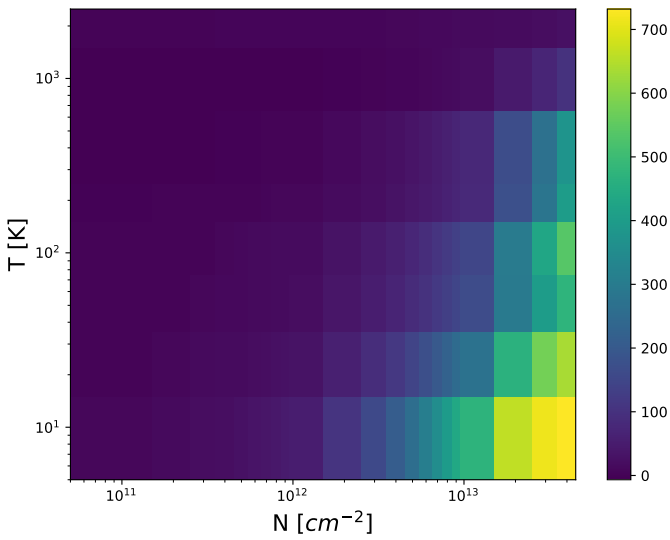

Fig. 4. Confidence intervals for the signal to noise of CN spectra using HARPS spectra of Beta Pictoris.

(Rossum 1995) and the open-source `Python` packages `numpy` (van der Walt et al. 2011), `matplotlib` (Hunter 2007), `astropy` (Astropy Collaboration et al. 2013). An online repository with materials used in this work is available at `https://github.com/sebastian-zieba/betaPic_comet`

References

A'Hearn, M. F., Millis, R. C., Schleicher, D. O., Osip, D. J., & Birch, P. V. 1995, Icarus, 118, 223
Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, A&A, 558, A33
Cordiner, M. A., Remijan, A. J., Boissier, J., et al. 2014, ApJ, 792, L2
Cremonese, G., Boehnhardt, H., Crovisier, J., et al. 1997, ApJ, 490, L199
Cremonese, G., Huebner, W. F., Rauer, H., & Boice, D. C. 2002, Advances in Space Research, 29, 1187
Dufay, J., Swings, P., & Fehrenbach, C. 1965, ApJ, 142, 1698
Fitzsimmons, A. & Cartwright, I. M. 1996, MNRAS, 278, L37
Fray, N., Bénilan, Y., Cottin, H., Gazeau, M. C., & Crovisier, J. 2005, Planet. Space Sci., 53, 1243
Fulle, M., Leblanc, F., Harrison, R. A., et al. 2007, ApJ, 661, L93
Hunter, J. D. 2007, Computing In Science & Engineering, 9, 90
Hyland, M. G., Fitzsimmons, A., & Snodgrass, C. 2019, MNRAS, 484, 1347
Lecavelier des Etangs, A., Cros, L., Hébrard, G., et al. 2022, Scientific Reports, 12, 5855
Pavlenko, Y., Kulyk, I., Shubina, O., et al. 2022, A&A, 660, A49
Preston, G. W. 1967, ApJ, 147, 718
Rappaport, S., Vanderburg, A., Jacobs, T., et al. 2018, MNRAS, 474, 1453
Rossum, G. 1995, Python Reference Manual, Tech. rep., Centrum voor Wiskunde en Informatica (CWI), Amsterdam, The Netherlands, The Netherlands
Schleicher, D. G. 2010, AJ, 140, 973
Slaughter, C. D. 1969, AJ, 74, 929
Thackeray, A. D., Feast, M. W., & Warner, B. 1966, ApJ, 143, 276
van der Walt, S., Colbert, S. C., & Varoquaux, G. 2011, Computing in Science & Engineering, 13, 22
van Sluijs, L., de Mooij, E., Kenworthy, M., et al. 2019, A&A, 626, A97
Zieba, S., Zwintz, K., Kenworthy, M. A., & Kennedy, G. M. 2019, A&A, 625, L13